 Name _____ Date _____

Superbug Showdown

What happens when bacteria evolve resistance — and how do we stop it?

Run the Superbug Showdown simulation from your unit page. Treat a patient with antibiotics and watch what happens to the bacteria population when doses are skipped or stopped early.

PART A — Before You Simulate — Predictions

1. Predict: if a patient takes antibiotics correctly for the full 10 days, what will happen to the bacteria population?

2. Predict: what will happen if the patient stops taking antibiotics after 3 days because they feel better?

PART B — Run the Simulation — Collect Data

Instructions: *Run THREE rounds using different treatment patterns. Record your results below. The sim tracks resistant vs. non-resistant bacteria separately.*

	Round 1 (Full course)	Round 2 (Stop at day 3)	Round 3 (Skip every other dose)
Day treatment ended	10	3	
Bacteria remaining at end			
Resistant bacteria at end			
Patient: recovered? (Y/N)			
Resistance evolved? (Y/N)			

PART C — Graph Your Results

Create a bar graph comparing the number of resistant bacteria remaining at the end of each round. Label your axes and give your graph a title.

[Draw your bar graph here]

PART D — Claim · Evidence · Reasoning (CER)

Question: Does finishing your full course of antibiotics matter for preventing antibiotic resistance?

3. CLAIM — Write a one-sentence answer to the question.

4. EVIDENCE — Use at least TWO specific numbers from your data table.

5. REASONING — Explain WHY your evidence supports your claim. Use the words: natural selection, mutation, resistant, reproduce.

PART E — Real World Connection

> ***Fact:*** *In 2021, 4.7 million deaths worldwide were linked to antibiotic-resistant bacteria. MRSA — a superbug staph bacterium — spreads through shared towels, gear, and turf burns. It is especially common among athletes.*

6. You get a skin infection from a turf burn at volleyball practice. The doctor prescribes 7 days of antibiotics. After 4 days you feel 100% better. Using what you learned today, explain why you should or should not stop taking the medication.

■ **Exit Ticket** Explain antibiotic resistance to someone who has never heard of it. Use an analogy (compare it to something familiar — like a sports team, a game, or something at school).

Adapted for 7th grade from New Visions for Public Schools, Humans vs. Bacteria (Biology Unit 2), CC BY-NC-SA 4.0. Simulation: yscabarney.github.io/MS_SCI_ADV_26/superbug-sim.html · Unit page: yscabarney.github.io/MS_SCI_ADV_26/humans-vs-bacteria.html